

MELITA, MB, Canada

WMO: 714470

Lat: 49.2812N

Lon: 100.9862W

Elev: 1463

StdP: 13.94

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-25.6	-20.7	-31.6	1.0	-25.4	-26.5	1.3	-20.6	28.7	10.0	25.6	12.3	6.3	260	0.677

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	21.4	86.4	68.5	83.5	67.2	80.5	65.8	72.8	81.5	70.6	79.3	68.3	76.7	10.8	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
70.1	116.9	77.6	67.5	106.8	75.4	65.3	98.7	73.2	37.5	81.6	35.5	79.1	33.5	76.5	83.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%		2.5%	5%	Mean		Standard Deviation	n=5 years		n=10 years		n=20 years		n=50 years			
Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Max
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	(p)
(4)	25.2	22.1	19.6	DB	-32.5	93.1	5.0	3.5	-36.1	95.6	-39.0	97.7	-41.8	99.6	-45.5	102.2
(5)				WB	-32.7	76.8	5.0	3.2	-36.3	79.1	-39.2	81.0	-42.0	82.8	-45.7	85.2

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	38.1	5.4	8.8	22.2	39.8	52.2	61.9	66.8	65.3	56.0	41.5	24.9	10.7	
	DBStd	24.47	15.35	14.34	14.09	10.72	8.47	6.52	5.40	6.20	8.15	9.36	12.08	14.01	
	HDD50	6104	1381	1153	864	332	76	3	0	0	33	291	754	1217	
	HDD65	10056	1846	1573	1328	757	405	132	44	73	285	728	1203	1682	
	CDD50	1766	0	0	1	25	143	360	520	475	214	28	0	0	
	CDD65	243	0	0	0	0	7	39	99	82	16	0	0	0	
	CDH74	2823	0	0	0	11	145	424	952	995	279	17	0	0	
	CDH80	747	0	0	0	1	35	90	229	319	70	3	0	0	
Wind	WSAvg	9.3	9.3	9.4	9.9	10.5	10.8	9.1	7.7	8.0	8.9	9.8	9.5	9.1	
Precipitation	PrecAvg	17.8	0.7	0.6	0.8	1.2	2.3	3.3	2.7	2.3	1.6	1.1	0.7	0.8	
	PrecMax	22.7	1.7	1.9	2.5	2.8	7.3	9.3	5.6	7.3	3.6	3.1	2.9	1.6	
	PrecMin	10.9	0.3	0.2	0.0	0.4	0.5	1.1	0.6	0.7	0.5	0.0	0.1	0.1	
	PrecStd	3.7	0.4	0.4	0.5	0.6	1.7	1.9	1.1	1.4	0.8	0.8	0.6	0.5	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	38.9	40.9	60.0	74.6	84.8	87.3	88.6	92.3	86.3	75.3	57.4	42.7	
		MCWB	35.2	37.1	49.3	55.0	61.2	69.3	72.1	68.9	65.7	57.2	47.2	37.5	
	2%	DB	35.4	36.1	50.3	67.9	77.8	82.3	85.5	87.2	81.4	67.3	50.2	36.8	
		MCWB	33.2	33.7	43.3	51.5	57.7	64.9	70.9	67.9	63.3	53.3	42.2	34.0	
	5%	DB	31.3	32.9	42.9	63.1	72.9	78.8	82.5	83.4	76.7	61.7	44.6	32.9	
		MCWB	29.7	31.7	39.0	48.7	55.5	64.4	69.9	66.7	61.3	50.5	39.2	30.9	
	10%	DB	27.4	29.3	38.3	57.4	68.5	75.5	79.5	79.8	71.4	56.2	39.7	29.1	
		MCWB	26.3	28.2	35.7	46.6	53.8	62.7	67.4	65.1	58.8	48.0	36.3	27.8	
	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	36.1	37.8	49.8	56.9	64.8	74.6	77.3	75.1	70.6	59.3	48.0	38.1
			MCDB	38.3	40.4	58.3	68.6	76.9	82.7	83.3	85.4	79.8	70.8	56.1	42.1
2%		WB	33.1	34.0	43.5	53.1	61.0	70.1	73.6	71.3	66.1	55.3	43.3	34.2	
		MCDB	35.2	35.7	50.2	65.2	71.1	77.2	81.5	81.7	76.4	64.3	49.1	36.5	
5%		WB	29.8	31.8	39.2	50.1	58.3	66.6	71.4	69.1	63.3	52.0	39.6	31.0	
		MCDB	31.3	33.0	43.0	60.3	68.8	74.7	80.0	78.9	73.5	59.7	44.0	32.9	
10%		WB	26.2	28.2	35.9	47.3	55.9	64.3	69.2	67.0	60.4	48.8	36.5	27.8	
		MCDB	27.2	29.4	38.1	56.4	65.2	72.4	77.1	76.5	69.0	55.9	39.7	29.2	
Mean Daily Temperature Range	5% DB	MDBR	17.5	18.0	17.2	21.1	22.7	20.1	21.4	24.2	23.2	20.0	16.6	16.0	
		MCDBR	18.8	17.0	21.5	30.1	29.9	25.5	25.4	29.8	31.5	29.9	23.9	19.2	
		MCWBR	17.1	14.9	15.4	15.7	14.0	12.8	13.2	13.8	15.8	17.2	16.6	16.7	
	5% WB	MCDBR	19.1	16.4	20.6	27.6	24.3	21.3	23.0	24.5	26.8	27.5	22.7	19.4	
		MCWBR	17.3	14.5	14.9	15.8	12.9	12.7	13.4	13.1	15.0	16.3	16.2	17.0	
Clear-Sky Solar Irradiance	taub	0.253	0.257	0.304	0.335	0.350	0.365	0.379	0.387	0.350	0.295	0.263	0.238		
	taud	2.332	2.348	2.377	2.385	2.393	2.395	2.370	2.326	2.402	2.533	2.489	2.410		
	Ebn at Noon	259	288	289	290	288	284	277	269	269	272	258	253		
	Edh at Noon	21	27	31	35	36	36	36	36	30	22	18	17		
All-Sky Solar Radiation	RadAvg	475	841	1236	1583	1833	1931	2067	1678	1240	760	441	339		
	RadStd	39	73	97	113	128	132	92	126	98	87	61	36		

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (15 neighbors)	N/A	N/A	N/A	N/A	N/A	-1.01	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page